

SAMRAT MAJHI

AI/ML Engineer | Agentic AI Developer

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Portfolio: samrat-portfolio.example.com | GitHub: github.com/samrat | LinkedIn: linkedin.com/in/samrat

SUMMARY

AI/ML Engineer specializing in agentic AI systems. Architected a 14-agent LangGraph healthcare assistant handling 100+ concurrent conversations with 100% grounded Retrieval-Augmented Generation (RAG) responses, and a banking intent classification system with 94% accuracy. Final-year B.Tech Computer Science Engineering student skilled in Python, LangChain, LangGraph, FastAPI, Docker, AWS, and GCP.

SKILLS

Languages & Libraries: Python, JavaScript, SQL, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn
Machine Learning & Data Science: Data Cleaning, Model Training, Hyperparameter Tuning, Data Pipeline Development, Exploratory Data Analysis (EDA), A/B Testing

GenAI & Deep Learning: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Hugging Face, Vector Databases, Large Language Models (LLMs), RNN, CNN, LSTM, Neural Networks

MLOps & Deployment: Docker, FastAPI, AWS, GCP, CI/CD, MLflow, Git, Apache Airflow

PROJECTS

ArogyaAI – Multilingual WhatsApp Multi-Agent Healthcare Assistant

Technologies: Python, JavaScript, CSS, FastAPI, LangGraph, RAG, Twilio, Docker, AWS EC2, Vector Database

Live App: arogyaai-demo.example.com | Code: github.com/samrat/arogyaaai

- Architected and deployed a 14-agent LangGraph healthcare assistant on AWS EC2, handling 100+ concurrent conversations across symptom triage, follow-up, disease guidance, and emergency escalation via Twilio WhatsApp
- Engineered a Retrieval-Augmented Generation (RAG) pipeline over a curated vaccination knowledge base, ensuring 100% of responses stayed grounded in verified medical source data (zero fabricated facts), delivering context-aware multilingual guidance to users
- Designed an end-to-end MLOps architecture using Docker, FastAPI, AWS EC2, and Vercel, exposing scalable REST APIs and enabling continuous deployment of a production-ready AI healthcare platform

User Intent Detection & Drift Monitoring System

Technologies: Python, Scikit-learn, TF-IDF, Logistic Regression, FastAPI, Render, Docker

Live App: intent-drift-demo.example.com | Code: github.com/samrat/intent-drift-monitor

- Automated banking query routing with 94% intent classification accuracy, cutting manual triage effort by 30%, by building a TF-IDF plus Logistic Regression model across balance, loan, fraud, and card queries for downstream NLP workflows
- Reduced misrouted requests by 25% by engineering a confidence-based drift detection framework that flags low-confidence predictions (below 0.75) for early model degradation detection
- Built and deployed a production-ready MLOps pipeline with FastAPI, Streamlit, Docker, Apache Airflow, and GitHub Actions CI/CD, enabling automated retraining, live monitoring, and self-service issue resolution

WORK EXPERIENCE

Pinnacle Labs Pvt Ltd – Data Science Intern

07/2025 – 08/2025, Remote

- Trained spaCy and machine learning classification models for sentiment analysis and fake news detection, achieving 89% accuracy on validation datasets across multiple model comparisons
- Built a CNN-based image classification pipeline using TensorFlow, with end-to-end data preprocessing, cleaning, and model evaluation, deployed on GCP

Quirky Connections Pvt Ltd – Generative AI Developer Intern

03/2025 – 04/2025, Remote

- Built and deployed two production-ready chatbots with PDF summarization and voice support using GPT-4 and Gemini APIs, achieving sub-2-second response time on AWS EC2
- Containerized applications with Docker and served via FastAPI REST backend on AWS EC2 for reliable production deployment

EDUCATION

Bachelor of Technology in Computer Science Engineering – Pailan College of Management & Technology (MAKAUT) 09/2023 – Expected 07/2026

CGPA: 8.6 | Kolkata, India

Diploma in Computer Science and Technology – Budge Budge Institute of Technology (WBSC TVESD) 08/2020 – 08/2023

CGPA: 7.4 | Kolkata, India

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Data Science Professional | AWS Certified Machine Learning – Specialty (MLS-C01)

COMPETITIONS & HACKATHONS

- Amazon ML Challenge – 2025:** Developed a smart product pricing model by processing 75,000+ product images and training deep learning models on Google Colab for large-scale data handling and optimization
- Google Gen AI Hackathon – 2025:** Built a generative AI solution to support and improve youth mental wellness